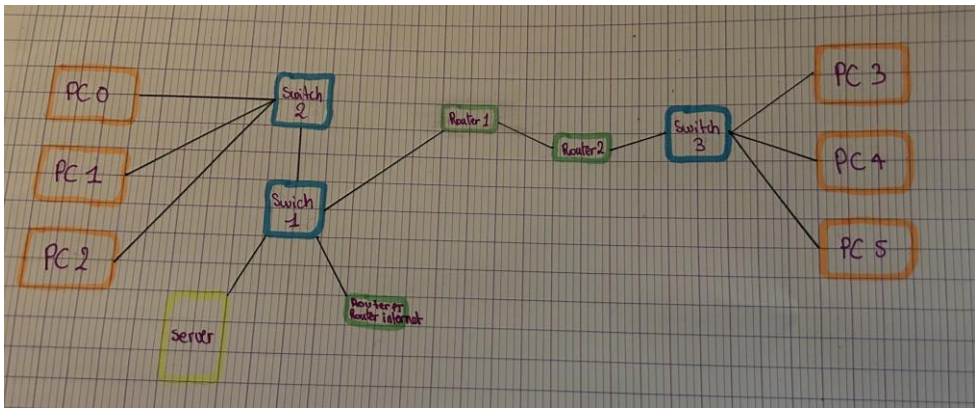
TP1- Mise en situation : Infrastructure réseau d'une PME multisite

1)

Etape 1:Topologie réseau:

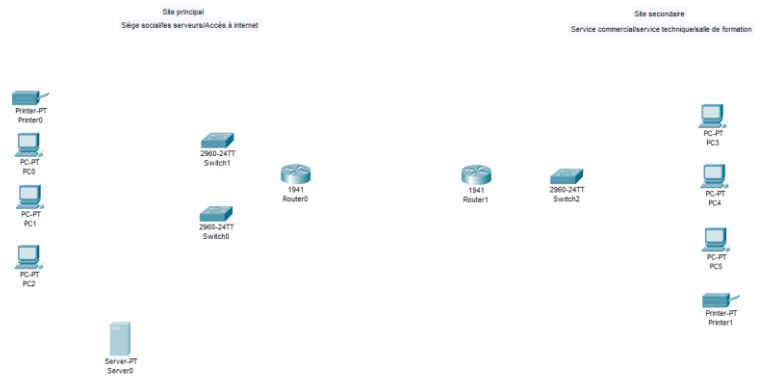


Etape 2:Plan d'adressage IP:

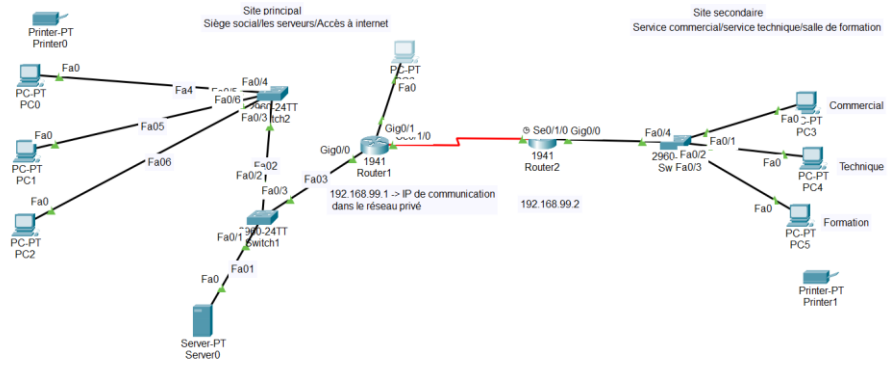
Étape 1 : Plan d'adressage IP

VLAN	Usage	Plage d'adresses IP	Masque	Passerelle
10	Département commercial	192.168.10.0/24	255.255.255.0	192.168.10.1
20	Département technique	192.168.20.0/24	255.255.255.0	192.168.20.1
30	Salle de formation	192.168.30.0/24	255.255.255.0	192.168.30.1
99	VLAN de gestion	192.168.99.0/24	255.255.255.0	192.168.99.1
WAN	Liaison inter-sites	10.0.0.0/30	255.255.255.252	10.0.0.1 (R1), 10.0.0.2 (R2)
40	Administration	192.168.40.0/24	255.255.255.0	192.168.40.1
50	Siege	192.168.50.0/24	255.255.255.0	192.168.50.1

Etape 3: Mise en place des différents équipements



Etape 4: Mise en place des connexions



2)Création des différents VLAN

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 60
Switch(config-vlan)#name
```

#enable

#configure terminal

#vlan ‘numéro’

#name ‘nom’

#exit

Exemple pour ajout VLAN 40 et 50 pour services administration et siège sur Switch 1,2 et 3

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 40
Switch(config-vlan)#name Administration
Switch(config-vlan)#exit
Switch(config)#vlan 50
Switch(config-vlan)#name Siege
Switch(config-vlan)#exit
Switch(config)#exit
```

Résultat pour # show vlan brief :

```
10   Commercial                       active
20   Technique                        active
30   Formation                        active
40   Administration                   active
50   Siege                            active
99   Gestion                          active
```

3)Configuration des ports/VLAN définis des Switch

Switch 3 --> Site secondaire (Service commercial/technique et salle de formation)

Port Fa0/1 - Commercial

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

```
10   Commercial                          active    Fa0/1
```

Port Fa0/2 - Technique

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

```
10   Commercial                          active    Fa0/1
20   Technique                           active    Fa0/2
```

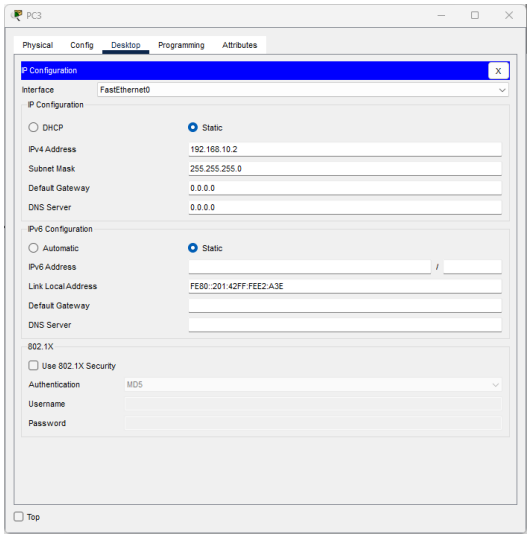
Port Fa0/3 - Salle de formation

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

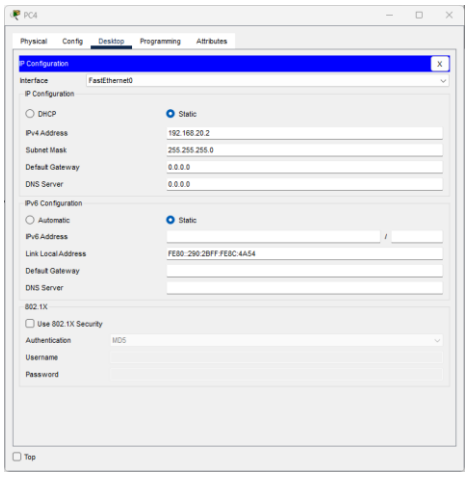
```
10   Commercial                          active    Fa0/1
20   Technique                           active    Fa0/2
30   Formation                           active    Fa0/3
```

Attribution des adresses IP et passerelles des PC du Switch 3

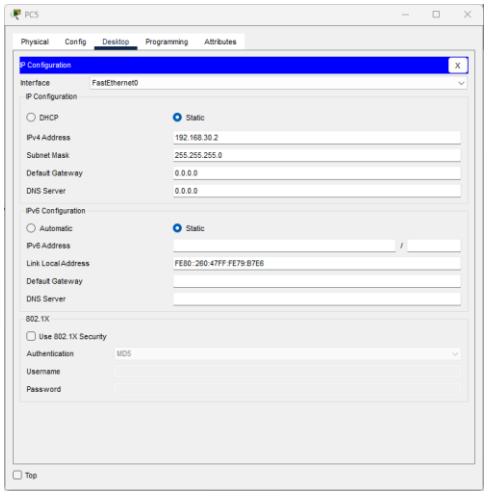
PC3 - Commercial



PC4 – Technique



PC5 – Salle de formation



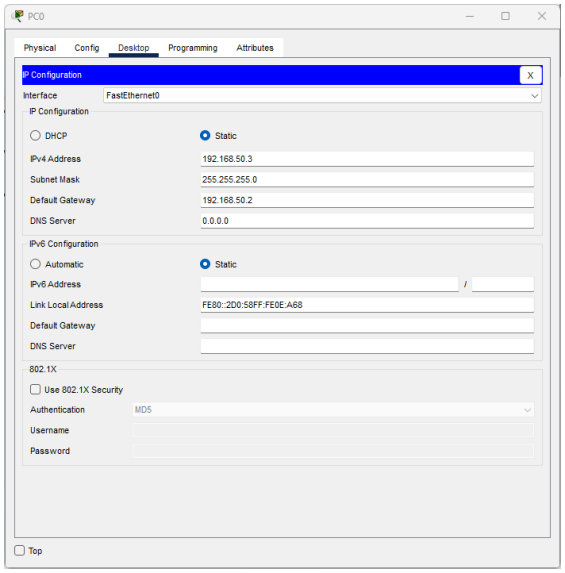
Switch 2 --> Siège social (VLAN 50)

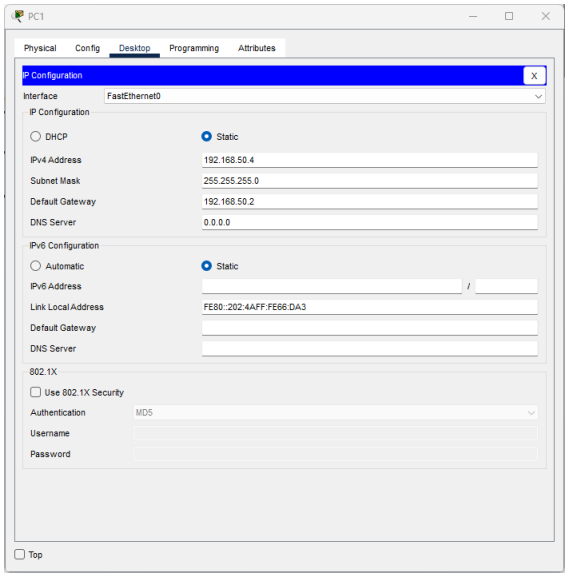
```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 50
Switch(config-if)#exit
Switch(config)#interface Fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 50
Switch(config-if)#exit
Switch(config)#interface Fa0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 50
Switch(config-if)#exit
Switch(config)#exit
```

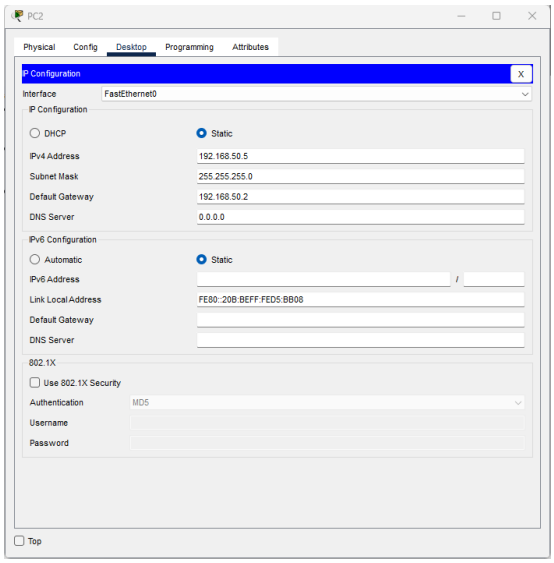
Switch 1 --> Administration (VLAN 40)

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#exit
```

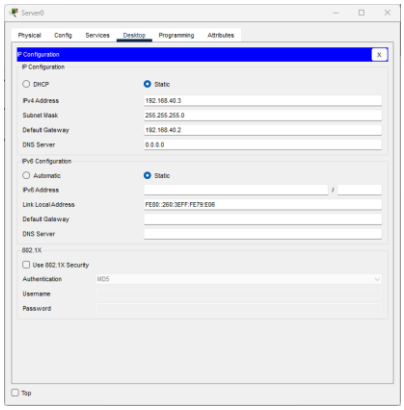
Attribution des adresses IP et passerelles des PC du Switch 2







Attribution adresse IP et passerelle du serveur



Configuration des trunk des Switch 1 et 2

Switch 1:

```
Switch(config)#interface Fa0/3
Switch(config-if)#switch mode trunk
Switch(config-if)#exit
Switch(config)#exit

Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/3       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/3       1-1005

Port        Vlans allowed and active in management domain
Fa0/3       1,10,20,30,40,50,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/3       1,10,20,30,40,50,99

Switch>
```

Switch 2:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/3
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
exit
```

```
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/3       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/3       1-1005

Port        Vlans allowed and active in management domain
Fa0/3       1,10,20,30,40,50,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/3       none

Switch>
```

Configuration du trunk entre Switch1 et Routeur1

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
```

4. Configurer le routage inter-VLAN

Configuration des sous-interfaces du Routeur1 (VLAN 40,50)

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES NVRAM  up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/0.50  192.168.50.1    YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    down
Serial0/1/0            10.0.0.2        YES manual up                    up
Serial0/1/1            unassigned      YES NVRAM  down                  down
Vlan1                  unassigned      YES NVRAM  up                    down
```

Ping entre PC1 et PC2 (VLAN similaire) et PC1 et Serveur (VLAN différent)

```
C:\>ping 192.168.40.3

Pinging 192.168.40.3 with 32 bytes of data:

Reply from 192.168.40.3: bytes=32 time=2ms TTL=127
Reply from 192.168.40.3: bytes=32 time<1ms TTL=127
Reply from 192.168.40.3: bytes=32 time<1ms TTL=127
Reply from 192.168.40.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.50.5

Pinging 192.168.50.5 with 32 bytes of data:

Reply from 192.168.50.5: bytes=32 time<1ms TTL=128
Reply from 192.168.50.5: bytes=32 time=6ms TTL=128
Reply from 192.168.50.5: bytes=32 time<1ms TTL=128
Reply from 192.168.50.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.50.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 1ms
```

5) Configurer le routage entre les deux routeurs

Configurer les routes

show ip route

Router1:

```
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/0
L       10.0.0.2/32 is directly connected, Serial0/1/0
S    192.168.10.0/24 [1/0] via 10.0.0.3
S    192.168.20.0/24 [1/0] via 10.0.0.3
S    192.168.30.0/24 [1/0] via 10.0.0.3
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L       192.168.50.1/32 is directly connected, GigabitEthernet0/0.50
```

Router2

```
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Serial0/1/0
L       10.0.0.3/32 is directly connected, Serial0/1/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
S    192.168.40.0/24 [1/0] via 10.0.0.2
S    192.168.50.0/24 [1/0] via 10.0.0.2
```

Ping entre PC0 (VLAN 50) et PC3 (VLAN 10) --> VLAN + Site différents

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=10ms TTL=126
Reply from 192.168.10.2: bytes=32 time=10ms TTL=126
Reply from 192.168.10.2: bytes=32 time=1ms TTL=126
Reply from 192.168.10.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 5ms
```

6) Assurer la connectivité Internet